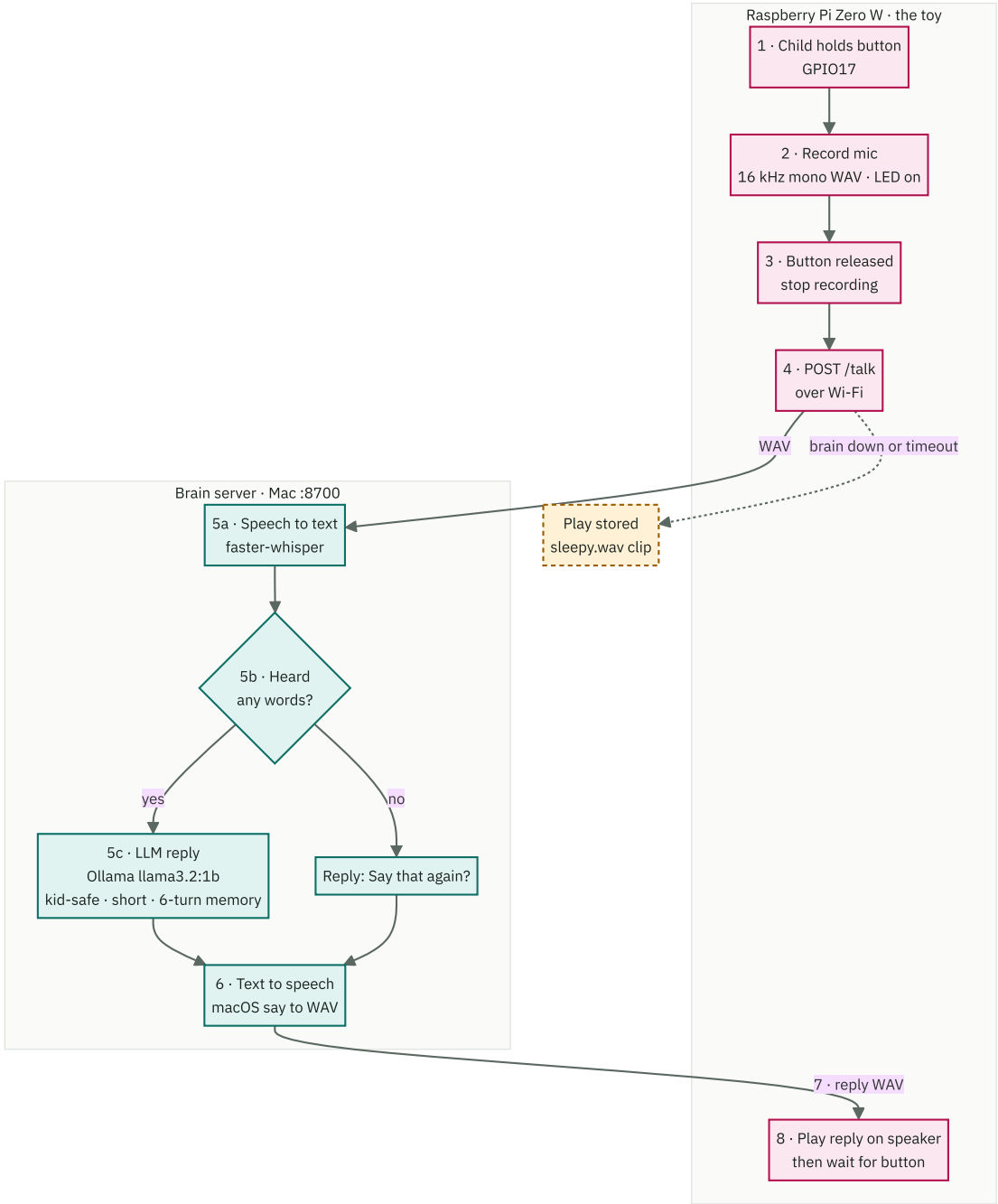
~/TALKTOY · ARCHITECTURE

A Pi Zero W that listens, and a Mac that thinks

The Pi Zero W has one 1 GHz core and 512 MB of RAM, so it can't run speech recognition or an LLM. The toy only records and plays sound. A brain server on the Mac, on the same Wi-Fi, turns speech into a reply.

One conversation, end to end

Runs on the Pi (the toy) · Runs on the Mac (the brain) · Failure path



Also running: the Pi's web page on :8080 (status, volume, talk button) and the Mac dashboard on :8700 (every chat logged). Both are for the parent's phone or browser.

Where the seconds go

Record `as long as the button is held`
arecord writes a 16 kHz mono WAV, about 32 KB per second.

Upload `~0.3 s`
A 4-second clip is about 128 KB over 2.4 GHz Wi-Fi.

Speech to text `~1–2 s`
faster-whisper `base.en` on the Mac's i5 CPU.

LLM reply `~2–4 s`
Capped near 60 words so the answer stays short.

Text to speech `~0.5 s`
macOS `say`, built in, nothing to install.

Play `length of the answer`
aplay on the speaker. Estimated wait after letting go: about 4–7 s.

Wiring on the Pi

PIN	CONNECTS TO
`GPIO17 · pin 11`	Push button, other leg to GND (pin 9)
`GPIO27 · pin 13`	LED through a 330 Ω resistor to GND
`GPIO18 · pin 12`	I2S BCLK: mic SCK and amp BCLK `I2S build`
`GPIO19 · pin 35`	I2S LRCLK: mic WS and amp LRC `I2S build`
`GPIO20 · pin 38`	I2S data in from INMP441 SD `I2S build`
`GPIO21 · pin 40`	I2S data out to MAX98357A DIN `I2S build`
`micro-USB OTG`	USB sound card instead of I2S `easy build`

Parts

Board	Raspberry Pi Zero W. A Zero 2 W is the same size and about 4× faster.
Audio, easy	USB sound card dongle + micro-USB OTG adapter
Audio, compact	INMP441 I2S mic + MAX98357A I2S amp, `dtoverlay=googlevoicehat-soundcard`
Speaker	3 W, 4–8 Ω
Input	Momentary push button, optional LED
Power	5 V 2 A USB, or a LiPo cell + boost board for a cordless toy

Web servers and endpoints

ADDRESS	WHAT IT DOES
`POST :8700/talk`	WAV in, spoken reply WAV out `Mac`
`GET :8700/`	Dashboard of every conversation `Mac`
`GET :8700/health`	Is the brain up and the model loaded `Mac`
`GET :8080/`	Toy status, last chat, volume `Pi`
`POST :8080/talk`	Record 5 s from a phone tap `Pi`

Kid safety lives in the brain. The persona prompt keeps answers short, gentle and age-appropriate, and every exchange is logged on the dashboard so a parent can read it. A small local model can still say odd things, so read the log during the first week.